

Mengchen Fan

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RESEARCH INTERESTS

- My research develops AI methodologies for learning from complex and heterogeneous data. I focus on efficient and reliable models, including model compression and adaptation, multimodal data fusion, longitudinal-data reasoning, and federated and decentralized learning. Medical imaging is a primary application area for my research.

EDUCATION

- **University of Alabama at Birmingham** *Aug 2022 – Present*
Ph.D. in Computer Science, GPA: 4.0/4.0 Birmingham, AL, United States
- **Syracuse University** *Sept 2017 – May 2019*
M.S. in Computer Engineering, GPA: 3.867/4.0 Syracuse, NY, United States
- **North China Electric Power University** *Sept 2012 – May 2016*
B.S. in Electrical Engineering, GPA: 3.47/4.0 China

RESEARCH EXPERIENCE

- **University of Alabama at Birmingham** *Aug 2022 – Present*
Doctoral Researcher in Computer Science Birmingham, AL, United States
 - Developed efficient medical AI models and longitudinal-data benchmarks, including region- and context-aware distillation that produced sub-10M-parameter 3D segmenters while preserving downstream performance and evidence-grounded temporal reasoning over complex patient trajectories.
 - Developed federated and decentralized learning methods to address client heterogeneity through personalized training, representative knowledge sharing, and efficient cross-client collaboration.
 - **Outcomes:** One MICCAI 2026 paper received an Early Accept (top 9% among 4,601 submissions) and was selected for a Spotlight presentation; one paper was published in the Q1 journal *Intelligent Systems with Applications* (ISWA).
- **Pulin Che Laboratory, UAB Heersink School of Medicine** *May 15 – Aug 15, 2026*
Research Intern Birmingham, AL, United States
 - Applied deep learning to ImageStreamX imaging-flow-cytometry data to analyze pathway-specific Golgi morphology, developing interpretable classifiers for inflammatory activation, signaling-pathway identification, and generalization across species and unseen compounds.

PROFESSIONAL EXPERIENCE

- **Citi Group** *Dec 2020 – Aug 2022*
Java Full Stack Developer Shanghai, China
 - Led full-stack development of an SMA platform supporting Basel IV operational-risk calculations using Spring and Java on the backend and Angular and JavaScript on the frontend.
 - Designed and integrated Oracle database workflows while delivering internal requirements in both waterfall and agile development environments.
 - Mentored and led new team members contributing to internal web projects for the Shanghai organization.

INVITED TALK

- **WNAR/IMS/Graybill 2024 Conference** *June 12, 2024*
Colorado State University Fort Collins, CO, United States

TECHNICAL SKILLS

Programming: Python, C++, C, Java, C#, SQL, JavaScript
Frameworks and Tools: PyTorch, Spring, Angular, Oracle, HTML, CSS

PUBLICATIONS

J–Journal, C–Conference, P–Patent

- [C.1] **Mengchen Fan**, Fei Zhao, Chengcui Zhang, Shuya Feng, and Baocheng Geng. *On Optimal Modality Compression for Bandwidth Constrained Multimodal Fusion*. **IEEE International Conference on Information Reuse and Integration for Data Science (IRI)**, 2026.
- [C.2] **Mengchen Fan**, Baocheng Geng, Xi Xiao, Tianyang Wang, Siyuan Mei, Pulin Che, Xiaoqian Jiang, and Qizhen Lan. *Detail Consistent Stage-Wise Distillation for Efficient 3D MRI Segmentation*. **International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)**, 2026, **Early Accept (9%)**.
- [C.3] **Mengchen Fan**, Tianyun Zhang, and Baocheng Geng. *A Unified Framework for the Convergence and Weight Pruning in Federated Learning*. **ACM Multimedia Asia**, 2025.
- [J.1] E. Chesnokova, D. J. Barrett, B. Geng, **Mengchen Fan**, T. Kulkarni, and S. Rothenberg. *An AI Physical Exam Can Predict Lung Function*. **American Journal of Respiratory and Critical Care Medicine**, 2025, vol. 211, A5307.
- [J.2] Joseph Casey, Qianjiao Chen, **Mengchen Fan**, Baocheng Geng, Roman Shterenberg, Zhong Chen, and Keren Li. *Distributed Collaborative Learning with Representative Knowledge Sharing*. **Mathematics**, 2025, vol. 13, no. 6, 1004.
- [C.4] **Mengchen Fan**, Keren Li, Tianyun Zhang, Qing Tian, and Baocheng Geng. *PFedDST: Personalized Federated Learning with Decentralized Selection Training*. **IEEE International Joint Conference on Neural Networks (IJCNN)**, 2025.
- [J.3] **Mengchen Fan**, Tianyun Zhang, Xiaolong Ma, Jiacheng Guo, Zheng Zhan, Shanglin Zhou, Minghai Qin, Fei Sun, Yen-Kuang Chen, Caiwen Ding, Makan Fardad, and Yanzhi Wang. *A Unified DNN Weight Compression Framework Using Reweighted Optimization Methods*. **Intelligent Systems with Applications**, 2025, vol. 27.
- [J.4] Yusen Zhai, **Mengchen Fan**, Baocheng Geng, Xue Du, Scott Snyder, and Larrell Wilkinson. *Impact of Phased COVID-19 Vaccine Rollout on Anxiety and Depression among the U.S. Adult Population, January 2019–February 2023: A Population-Based Interrupted Time Series Analysis*. **The Lancet Regional Health – Americas**, 2024.
- [C.5] **Mengchen Fan**, Baocheng Geng, Keren Li, Xueqian Wang, and Pramod K. Varshney. *Interpretable Data Fusion for Distributed Learning: A Representative Approach via Gradient Matching*. **International Conference on Information Fusion (FUSION)**, 2024, pp. 1–8.
- [P.1] Jinju Qian, Zhichao Mo, Zhangquan Rao, Weibing Yue, Ke Wang, **Mengchen Fan**, Xiangyang Peng, Xiaofeng Pang, Lin Yi, and Rui Wang. *Routing Inspection Path Planning Method and Device for Routing Inspection Robot*. **Chinese Patent CN106156888B**, 2020.